1, Rings and ideals

Ring: $(R, +, \cdot)$
- $(R, +)$ abelian group
- $(R, \cdot)$ semigroup
- distributive law

All our rings will have a unit $1 \in R$
All our rings unless specified will be commutative

$0 \in R$ neutral element under $+$
The trivial ring $\{0\}$ has $1 = 0$
$u \in R$ is a unit (= invertible element) if there exist $\exists v \in R.\ uv = vu = 1$. This v is unique $v = u^{-1}$

$R^\times$ or R^{-1} or $GL(R)$... units of R

Ring homomorphism is a map $\varphi: R \to R'$ that preserves sums, product and ones

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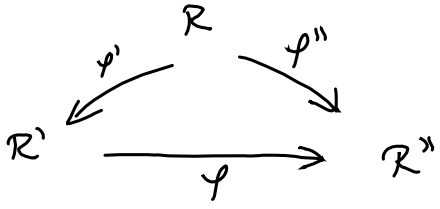
R-algebras: ring R' with a homomorphism

$\varphi : R \to R'$

called structure homomorphism

Every ring is a $\mathbb{Z}$-algebra

R-algebra homomorphism between R', R'' is a ring homomorphism that is compatible with structure maps



Examples:

- $\mathbb{Z}_2$
- R ring, Y set

  $R^Y = \{ Y \to R \}$ is a ring with point wise operations $f, g \in R^Y$ $(f+g)(y) = f(y) + g(y)$

  $(f \cdot g)(y) = f(y) \cdot g(y)$

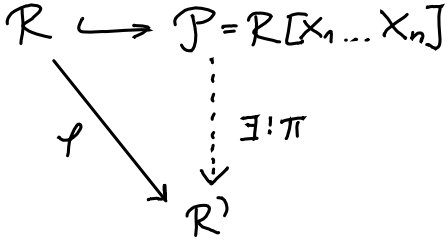
  identity: $y \mapsto 1$

Ring of polinomials: $\mathcal{P} = R[X_1, \dots X_n]$

$\mathcal{P}$ has the following universal property (UL):

$\forall$ hamamorphism $\varphi : R \to R'$. $\forall a_1, \dots a_n \in R$

$\exists!$ hamamorphism $\pi : \mathcal{P} \to R'$ s: $\pi_{|R} = \varphi \wedge \pi(X_i) = a_i$



$$\pi\left(\sum_i r_i X_1 \cdots X_n\right) = \sum_i \varphi(r_i) a_1 \cdots a_n$$

• Ideal: $I \trianglelefteq R$ $\quad 0 \in I$

$x, y \in I \Rightarrow x+y \in I$

$r \in R, x \in I \Rightarrow rx \in I$

$S \subseteq R$ $\quad \langle S \rangle$ ... Ideal of R generated by S

$\langle S \rangle = \{ \sum_{s \in S}^{\text{finite}} a_s s \; ; \; a_s \in R \}$

$\langle a \rangle$ ... principal ideals

Operations on ideals

• I_λ $(\lambda \in \Lambda)$; ideals of R

$$\sum_{\lambda \in \Lambda} I_\lambda = \{ \sum_{\lambda \in \Lambda}^{\text{finite}} b_\lambda \; ; \; b_\lambda \in I_\lambda \}$$

• $\cap I_\lambda \trianglelefteq R$ $\qquad \bigcap_{\emptyset} I_\lambda = R$

• $\prod I_\lambda = \langle \{ \prod_{\lambda \in \Lambda'} b_\lambda \; ; \; \Lambda' \overset{\text{finite}}{\subseteq} \Lambda, \; b_\lambda \in I_\lambda \} \rangle$

• $(I : J) = \{ x \in R \; ; \; xJ \subseteq I \}$ transporter

• $I \cdot J \subseteq I \cap J \subseteq I, J \subseteq I + J$

$I \subseteq (I : J)$

$I(J+K) = I \cdot J + IK$ DN

$I \trianglelefteq R$ is proper if $I \subsetneq R$

$I \trianglelefteq R$ is trivial if $I = \{0\}$

$\langle x \rangle = R \Leftrightarrow x \in R^\times$

($\Leftarrow$) obvs

($\Rightarrow$) $\langle x \rangle = R \Rightarrow \exists u \in R \quad ux = 1 \Rightarrow x \in R^\times$

$\varphi : R \to R' \qquad I \lhd R$

$\varphi(I)$ tipically not ideal (

$\mathbb{Z} \hookrightarrow \mathbb{Q}$

$\uparrow \quad \nearrow$

$2\mathbb{Z}$

$\langle \varphi(I) \rangle =: I^e$ "estention of I"

$I' \lhd R'$

$\varphi^*(I') \lhd R$ "kontraction of I'" $I'^c = \varphi^*(I')$

$\varphi : R \to R' \Rightarrow \ker \varphi = \varphi^*(0) \lhd R$

$\ker \varphi = \{0\} \Leftrightarrow \varphi$ is injective

$I \trianglelefteq R$ Form cosets $\{x+I ; x \in R\} = R/I$

$$(x+I)+(y+I) = (x+y)+I)$$
$$(x+I)(y+I) = xy+I$$

Quotient or factor ring

$$K : R \longrightarrow R/I \qquad x \mapsto x+I$$

canonical quotient homomorphism

K is surjective

$\ker K = I$

$$\boxed{R/\ker\varphi \cong \operatorname{im}\varphi}$$

$R \quad I \trianglelefteq R \quad I \subseteq J \trianglelefteq R \quad K : R \to R/I$

$$J/I = \{y+I \mid y \in J\} = K(J) \trianglelefteq R/I$$

maps $J \mapsto J/I$ in $J' \mapsto K^{-1}(J)$ are inverse hence bijection

$$\frac{R/I}{J/I} \cong R/J$$

Proposition: $\mathcal{P} = R[X]$; $a \in R$. Let $\pi: \mathcal{P} \to R$ be the R-ring homomorphism $X \mapsto a$

a) $\ker \pi = \{F \in \mathcal{P};\ F(a) = 0\} = \langle x - a \rangle$

b) $\mathcal{P}/\ker \pi \cong R$

Proof: $G := X - a$

$\forall F \in \mathcal{P}.\ \exists H \in \mathcal{P}.\ \exists r \in R\ \ni: F = GH + r$

B; linearity it suficies to prove this for $f = X^n$

$f = X^{n-1}X = X^{n-1}(G + a) = GX^{n-1} + aX^{n-1}$

Repeat this starting with X^{n-1}

a) $\pi(F) = \underset{0}{\pi(G)}\,\pi(H) + \underset{r}{\pi(r)} = r$

Hence $F \in \ker(F) \Leftrightarrow r = 0 \Leftrightarrow F(X-a) \cdot H$

$\Leftrightarrow F \in \langle X - a \rangle$

b) $\mathcal{P}/\ker \pi \cong \operatorname{im} \pi = R$

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$\deg F = \max \{\deg(M);\ M \text{ monomial in } F\}$

$M = X_1^{\alpha_1} \dots X_n^{\alpha_n}$

$\deg(M) = \sum \alpha_i$

$\deg(0) = -\infty$

$\deg(F \cdot G) \leq \deg F + \deg G$

Idempotents if $e^2 = e$

Poglej si pri Filipu, če je potrebno

Prop: Let R be a ring and e', e'' complementary idempotents ($e''=1-e'$). Set $R'=Re'$ in $R''=Re''$

Define $\varphi: R \longrightarrow R' \times R''$

$$x \longmapsto (xe', xe'')$$

Then R is isomorphiseen $R' \cong R/Re''$, $R'' \cong R/Re'$

Proof:

Consider $\varphi': R \longrightarrow R$

$$x \longmapsto xe'$$

φ' is homomorphism $\varphi'(xy) = xye' = xye'^2 = \varphi(x)\varphi(y)$

$\ker \varphi' = R''$

$(\subseteq)$ $\varphi'(x)=0 \Rightarrow xe'=0$

$$x \cdot 1 = x(e'+e'') = 0 + xe'' \in R''$$

$(\supseteq)$ $\varphi(xe'') = x\underbrace{e''e'}_{=0} = 0$

Then $\operatorname{im}\varphi = R'$ and $R' \cong R/Re''$

Similarly we have $\varphi'': x \longmapsto xe''$

Then $\varphi = (\varphi', \varphi'')$ is homomorphism

$\ker \varphi = 0$

$x \in \ker\varphi \Rightarrow xe'=0$ and $xe''=0$

$\Rightarrow 0 = x(e'+e'') = x \cdot 1 = x$ ✓

$\operatorname{im}\varphi = R' \times R''$

$(xe', x''e'') = \varphi(x'e' + x''e'')$

$\Rightarrow \varphi$ is isomorphism

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II Prime ideals

$x \in R - \{0\}$ is a zero devisor $\Leftrightarrow \exists y \in R . xy = 0$

$zdiv R = \{ x \in R \text{ zero devisor} \}$

Def $S \subseteq R$ is multiplicative if $1 \in S$ & $S \cdot S \subseteq S$

$P \trianglelefteq R$ is prime if $R - P$ is multiplicative

Eg: $P \trianglelefteq R$ is prime

Ring is a field if $0 \neq 1$ and an every non zero element is invertable

Ring where $\langle 0 \rangle$ is prime ideal is integral domain (od. helobar)

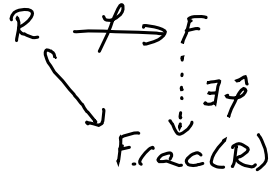
such ring has no zero devisors

- R integral domain $\Rightarrow$ can form a field of fractions

$$Frac(R) = \left\{ \frac{a}{b} ; a \in R ; b \in R - \{0\} \right\}$$

- every integral domain is a subring of a field
- every subring of a field is integral

UL for Frac(R): Each embeding of R into F Field extends to Frac(R) $\xrightarrow{\hat{\varphi}}$ F



$\hat{\varphi}\left(\frac{a}{b}\right) = \frac{\varphi(a)}{\varphi(b)}$

- Polynomial rings over integral domains

  R domain, $P = R[x]$

  P is also a domain and $\deg(FG) = \deg F + \deg G$

  We order P with so-called graded lexicographic (grlex) ordering:

  order first by degree, then lexicografically

  $x_1^4 x_2 < x_1^3 x_2^3 x_3 < x_1^2 x_2 x_3^4$

  Among all the monomials pick the largest one. This is LM(F) = leading monomial of F

  LT(F) = leading term of F $a \cdot LM(F)$ if $a \cdot LM(F)$ appears in F

  LC(F) = leading term of F a

$F, G \in \mathcal{P} \Rightarrow LT(FG) = LT(F) \cdot LT(G)$

$LT(F) = a \cdot m$, $LT(G) = a \cdot n$

Then $a \cdot b \neq 0$ bcos R is a domain

$\deg(m \cdot n) = \deg(m) + \deg(n) = \deg(F) + \deg(G)$

if $m' \neq m$ or $n' \neq n$ then $m'n' <_{grlex} mx$ by max of m, n !

Hence $LT(F \cdot G) = abmn = LT(F) \cdot LT(G)$

$\mathcal{P}$ domain $\Rightarrow$ hen form $Frac(\mathcal{P}) =: K(X)$,

$K = Frac(R)$ ↑ field of rational functions